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On the possibility of a widespread remagnetization of pre-Oligocene rocks from Northeast Japan and the Miocene rotational opening of the Japan Sea

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We present new paleomagnetic results from the well dated Miyako Cretaceous sediments (100–110 Ma) from Northeast Japan. These results, combined with those of Tosha [1], yield an in-situ characteristic direction $D = 321°$, $I = 54.5°$ ($\alpha_{95} = 4.5°$, $N = 14$ sites; reduced to a reference point at 40° N, 142° E). This direction is found to coincide with that of most older plutonic and sedimentary rocks of Devonian to lower Cretaceous age. It is also identical with the westerly pre-folding direction which is preserved in many Oligocene (20–40 Ma) formations from Northeast Japan [1,2]. In contrast, all recent formations (0–17 Ma) have been magnetized in the direction of the present axial dipole field. Only the Oligocene and Miocene results appear to be primary, or at least pre-folding. The Miyako sulfide-bearing sediments and lower Cretaceous (110–125 Ma) magnetite-bearing granites could either still bear a primary magnetization or be completely remagnetized by a low temperature chemical event. Evidence for such events is now found in many places, and as close as South Korea. Available data constrain the Oligo-Miocene history of Northeast Japan and indicate at least 20/30° counterclockwise rotation with respect to mainland Asia during the opening of the Sea of Japan. On the other hand, the pre-40 Ma history of Northeast Japan is not well constrained and three models are proposed which are compatible with various interpretations of the data. None of them can presently document pre-Oligocene motion of Northeast Japan with respect to Asia. The most "economical" model implies widespread remagnetization. We conclude that, because of the scarcity of well tested primary magnetization directions, the classical bending of the Japanese Islands rests on weaker grounds than generally realized and that no pre-40 Ma apparent polar wander path of the Japanese Islands can safely be proposed.

1. Introduction

Constraining the tectonic evolution of the Japanese Islands has been a challenge to paleomagneticians, particularly since Kawai et al. [3] first proposed that the main island of Honshu has been bent to form its well-known bow shape, sometime in the late Cretaceous. More recent approaches have linked this bending with subduction of the Pacific plate and back-arc spreading and opening of the Sea of Japan (Fig. 1). A large number of paleomagnetic papers has been devoted to the subject (some of them listed in Sasajima's [4] review), with various proposals as to the total angular amount of bending and age of the main motion. One of the most recent studies [2] concludes that the Sea of Japan opened by counterclockwise rotation of Northeast Japan through 50° about a pivot to the northeast and clockwise rotation of Southwest Japan through 54° about a pivot to the southwest, and that these events took place in a few million years in the Miocene. Recent analyses of onland structural features and of the physiography of the Japan Sea have led to conflicting models: Faure and Lalevée [5] argue for a bending model similar to that of Otofuji et al. [2], with somewhat reduced values for the rotation angles, whereas Lallemand and Jolivet [6] propose that the Japan Sea opened as a north-south pull-apart basin from the Eocene to the Lower Miocene.

Paleomagnetic data for periods prior to the opening of the Japan Sea are of great importance on their own to provide constraints on the (late Mesozoic and early Cenozoic) configuration and evolution of the Asian continent. Indeed, not many data are available for the huge continental assemblage that comprised North and South China, Korea, Siberia and parts of Japan prior to the opening of the Japan Sea (see e.g. [7]). Unfor-

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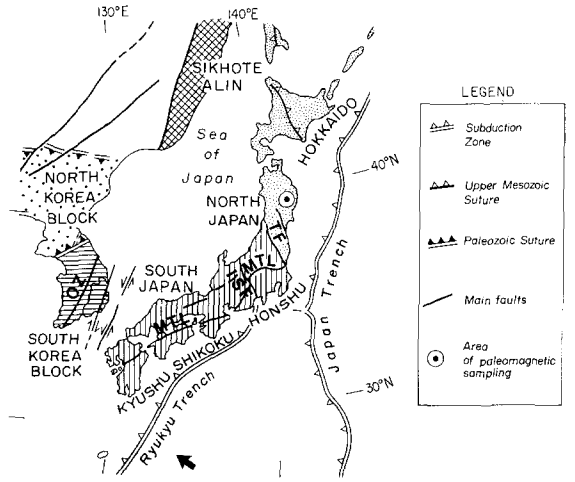


Fig. 1. Geodynamic setting of the Japanese Islands. The North Japan, South Japan, North Korea, South Korea and Sikhote Alin blocks are indicated by different patterns of shading. The Miyako paleomagnetic sampling area is indicated by a circle. Major faults include the Tanakura fault (*TF*), the Itoigawa-Shizuoka fault (*ISF*), the Median Tectonic Line (*MTL*) and the Ogchen zone (*OZ*).

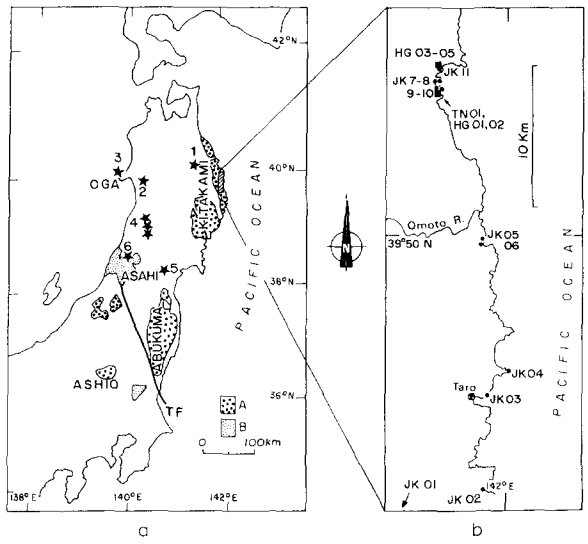


Fig. 2. Paleomagnetic sampling sites in northern Honshu, near to and north of the Tanakura fault (*TF*). Sites of sampling of the Miyako sediments are enlarged in Fig. 2b (JK sites are from this study, TN and HG sites from Tosha, [1]). Paleomagnetic sampling areas discussed in the text include Nonohe (*1*), Taihezan (*2*), Oga peninsula (*3*), Yuzawa (*4*), Zao (*5*) and North Asahi (*6*). Patterns: *A* = Cretaceous granitic rocks, *B* = Paleogene granitic rocks (redrawn after Otofuji et al. [2]).

tunately, most published paleomagnetic results from Cretaceous Northeast Japan have been obtained from plutonic bodies (Fig. 2): the Abukuma granites [8,9], the Kitakami granites [8,10,11] and more recently the Iide and Ashio granodiorites [2]. Sasajima [4] has shown that several early results did not pass minimum acceptability criteria; moreover, structural corrections are most of the time unattainable in plutonic bodies, preventing stability tests. The situation is particularly worrying since several recent studies have shown the importance of remagnetization in and around Japan, for instance in South Korea [7] or in the Kitakami massif in Northeast Japan [1].

For these reasons, we decided to try and obtain results from sedimentary formations, hopefully not too severely affected by tectonic or metamorphic events. Occurrences of such formations of Cretaceous age is extremely rare in Northeast Japan. A notable candidate (maybe the only one?) is the Miyako Group on the eastern coast of Japan (Fig. 2). This Group consists of small, almost monoclinal, exposures of greyish sandstones and siltstones, with a basal conglomerate. The Miyako sediments have been well dated with marine faunas, including ammonites, as being Aptian to Albian in age [12]. The Cretaceous sediments unconformably overlie the Kitakami granites, to the north of the Hayachine tectonic belt; these granites have been dated by the K-Ar method at 110–125 Ma [13]. The age of the Miyako Group is thus estimated to lie between 100 and 110 Ma. Tosha [1] recently sampled the Miyako Group at 9 sites, and obtained paleomagnetic results for six sites spread over less than 2 km. We have extended the sampling to an additional 10 sites, spread over approximately 25 km along the sea coast. Ninety cores were drilled from nine sites using a portable drill. Orientation was performed with a magnetic compass and, whenever possible, with a solar compass.

2. Paleomagnetic data

In each site, two pilot specimens were subjected to thermal demagnetization in steps of 40 °C and two other pilot specimens to alternating field (AF) demagnetization in steps of 5 mT. All magnetic measurements were done with a two-component CTF cryogenic magnetometer and included a cor-

rection (when necessary) for (very weak) spurious magnetization of the sample holder. The noise level of the magnetometer is estimated to be about (or less than) 10^{-11} A m^2 (corresponding to a magnetization of 10^{-6} A m^{-1} for a sample of volume 10 cm^3). The cryogenic magnetometer is particularly useful for the weakly magnetized sandstones which have typical natural remanent magnetizations of 10^{-3} to 10^{-4} A m^{-1} prior to demagnetization. On the spinner magnetometer used in Tosha's [1] study, a single measurement took up to two or three hours, compared to a few minutes on the cryogenic magnetometer.

A number of sample vector demagnetization diagrams are displayed in Fig. 3 (in geographic, i.e. in situ coordinates). The diagrams involve a number of common features, among which we can mention weak magnetization, very noisy behaviour upon demagnetization and remagnetization beyond moderate temperatures. Despite this, several components can be separately identified in the vector plots. A soft component, sometimes with a direction close to the present geomagnetic field direction in Japan, is removed by 15 mT (JK 07-07 A, Fig. 3f) to 40 mT (JK 01-04B, Fig. 3a) upon af cleaning, and up to 230 °C (JK 03-01A, Fig. 3b) upon thermal cleaning. In some cases, this component dominates the entire demagnetization spectrum. A second consistent component with northwesterly declination (A) appears next in most demagnetization diagrams. Its unblocking temperature spectrum ranges from a low of 100–230 °C to a high of 300–400 °C (Fig. 3b,c,g) and its coercivity spectrum from 5–40 mT (low) to 30–90 mT (high) (Fig. 3a,d,e,f). In some sites, another component with easterly declination is found between 200 and 300 °C: this component is found mostly in site JK 3, only in 4 out of 13 specimens and has not been interpreted because of its rare occurrence and lack of consistency. Be-

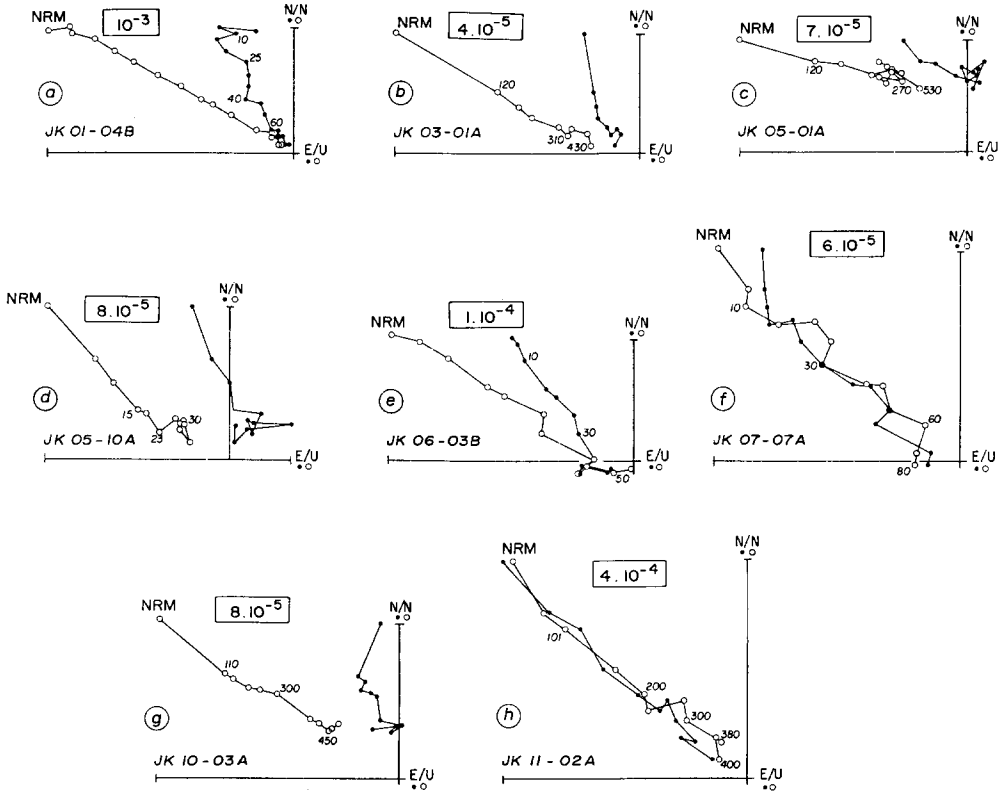


Fig. 3. Representative demagnetization vector diagrams from specimens of the Miyako sediments. The specimen identification is given near the abscissa axis. The value of NRM in A m^{-1} is given in a square. Closed symbols are for the horizontal projection and open symbols for projection in the NS vertical plane. (a), (d), (e) and (f) are alternating field demagnetizations (values in mT); (b), (c), (g) and (h) thermal demagnetizations (values in °C).

yond 300–400 °C, specimens show either no further demagnetization (e.g. JK 05-01A, Fig. 3c; JK 10-03A, Fig. 3g), or they become remagnetized, with large spurious remagnetizations (which have been omitted from the diagrams for clarity: such is the case mainly in sites JK 3 and JK 9, and JK 6 and JK 11 to a lesser extent). In those specimens for which there is almost no demagnetization taking place between 300 and 550 °C (e.g. Fig. 3c and g), the remaining component of magnetization appears to be close to the direction of the present field. On further heating, remagnetization occurs between 530 ° and 570 °C and it has not been possible to make any significant measurement at higher temperatures.

In order to complement af and thermal treatments, we have also studied, for one specimen from each site, the acquisition of isothermal remanence (IRM) in fields up to 2 T, and then the evolution of saturation IRM (IRM_s) as a function of temperature, in order to monitor changes in magnetic mineralogy. IRM curves (Fig. 4a) indicate that 75–96% of saturation is reached at room temperature in a field of 0.2 T, saturation being essentially complete by 0.4 T. In most samples, IRM_s increases significantly when the temperature reaches 300–400 °C (Fig. 4b): the increase is over one order of magnitude by 300 °C, and two orders of magnitude beyond 450 °C in sites JK 5 and JK 9. The increase is well over two orders of magnitude in site JK 2, from which we were unable to isolate any consistent direction (this site is actually omitted from further discussion). The IRM_s change is also very strong in site JK 8. The change is weaker in sites JK 3, 6, 7, 10 and 11, the increase reaching about 10 at temperatures of 400–500 °C. These experiments confirm the results from thermal demagnetizations, i.e. the occurrence of strong mineralogical changes beyond 300–400 °C in most cases, preventing use of directions at higher temperatures. There is a rather good correlation between the scatter in estimated directions from demagnetization diagrams (see values of confidence interval α_{95} in Table 1) and the intensity of the IRM_s change upon heating.

In his study of the same Miyako Group sandstones, Tosha [1] showed unblocking temperatures up to 450 °C and very good agreement of characteristic directions of magnetization upon AF treatment (up to 20 mT only). A thermomagnetic analysis indicated a Curie temperature of 580 °C [1, p. 52], suggesting that magnetite was the sole carrier of magnetic component A. Indeed, magnetite is observed in thin sections of samples from site JK 9, with a grain size of several tens of microns (M. Semet, personal communication, 1985). However, the noisy character of the demagnetization diagrams and the occurrence of chemical changes in the 300–400 °C range (with an increase in saturation IRM) indicate that iron

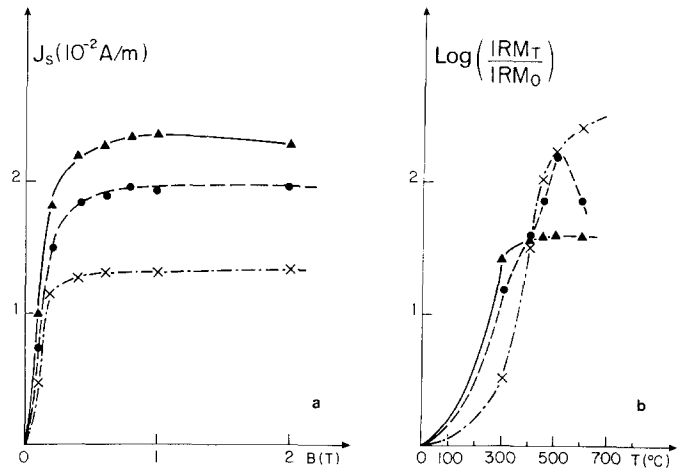


Fig. 4. (a) Acquisition curves of induced (isothermal) remanent magnetization (IRM) in magnetic inductions B up to 2 T. (b) Evolution of saturation IRM (normalized to its room-temperature value) as a function of reheating temperature. Symbols are as follows: X = site JK 05; ▲ = site JK 08; ● = site JK 09.

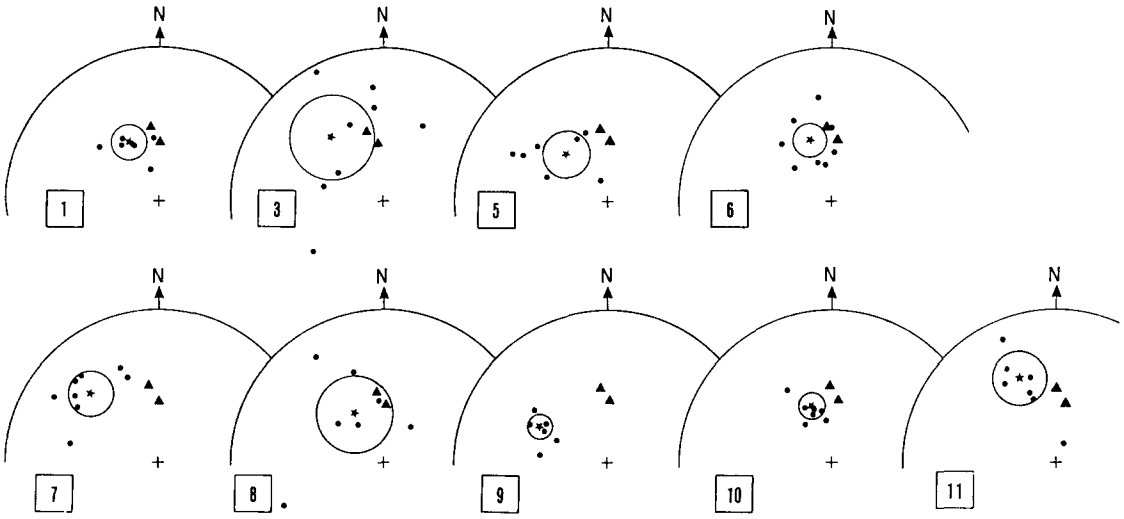


Fig. 5. Lower hemisphere equal-area projections of characteristic directions of magnetization for all demagnetized specimens of the Miyako sediments. The site number is given in the lower left-hand corner in a square (e.g. 1 = JK 01). The mean for each site is given as a star with its 95% confidence interval. The two triangles in each frame represent the present field direction and present dipole field direction in the sampling area.

sulfides might be essential carriers of magnetization in our samples. This is supported by observations under the optical microscope which reveal possibly primary pyrite and pyrrhotites of 10 μm size in site JK 9 and both primary and secondary pyrite in site JK 11 (M. Semet, personal communication, 1985). The fact that in several cases a consistent A component can still be isolated somewhat above 310 °C (the maximum Curie temperature of pyrrhotites) suggests that some magnetite must also be present, possibly in smaller amounts.

In summary, a rather consistent northwesterly magnetization A, possibly carried by iron sulfides and some magnetite, occurs in most Miyako samples, with an unblocking temperature range from 100–150 °C to 300–400 °C and is also found in AF demagnetization diagrams roughly between 10 and 40 mT. This component has been determined by principal component analysis and statistical results are summarized in Table 1 and Fig. 5; the precision angle is found to vary from 6 to 25°, depending on the site. Although the results are not of very high accuracy, they are quite consistent, and yield a mean direction $D = 319°$, $I = 53°$ ($\alpha_{95} = 7.5°$, $N = 8$ sites; Fig. 6-4). The mean hardly changes if the three sites with α_{95} close to or larger than 20° are discarded ($D = 316°$, $I = 54°$, $\alpha_{95} = 11.5°$, $N = 5$ sites). This is in excellent agreement with the mean obtained by Tosha [1] from six sites over a more restricted (less than 2 km) area: $D = 322°$, $I = 56°$, $\alpha_{95} = 7.5°$ (Fig. 6-3). The various means are all within the intersection of their 95% confidence intervals and are thus statistically undistinguishable (Fig. 6-6). They can be combined to yield an overall mean direction $D = 321°$, $I = 54°$ ($\alpha_{95} = 5°$, $N = 14$ sites; Table 1). The Miyako Group unfortunately occurs as an essentially monoclinal sequence, dipping about 15–30° to the northeast with an azimuth ranging from 0 to 30°) and does not allow one to perform a fold test. Yet, we should note that the direction prior to bedding correction is significantly different from the present or dipole field direction at the sampling locations and is thus likely to have retained information on a paleofield direction. After bedding correction, the direction becomes $D = 0°$, $I = 63.5°$ ($\alpha_{95} = 5.5°$). The confidence interval includes the dipole field direction and is not far from the present field direction, although this is inferred to be a coincidence.

3. Summary of available paleomagnetic results from North Japan

We will now attempt to critically review available data, not only from the Miyako sedimentary formation, but also from all other Mesozoic and Cenozoic formations in North Japan, including

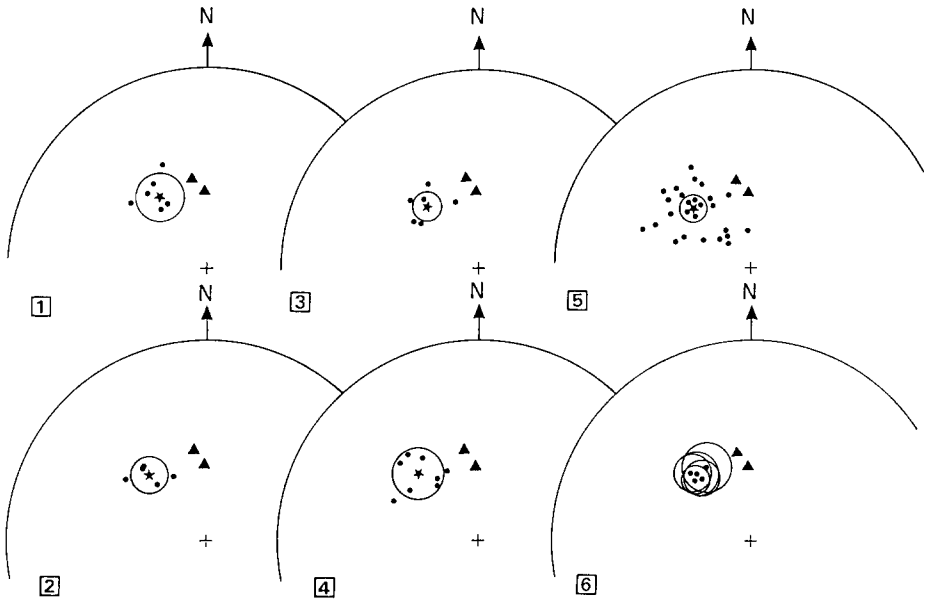


Fig. 6: A summary of paleomagnetic directions in Northern Honshu. (1) Devonian to Jurassic sediments from Tosha [1] (Table 2a); (2) Lower Cretaceous plutonic rocks (listed in Table 2b); (3) Miyako Cretaceous sediments from Tosha [1] (sites HG and TN in Table 1); (4) Miyako Cretaceous sediments from this study (sites JK in Table 1); (5) Direction for the 27 formations and studies listed in Table 3 for the period 20–40 Ma. Note that in 6-1 to 6-4 directions are in in-situ coordinates, whereas they have been corrected for bedding in 6-5. (6) Comparison of the mean directions from the five previous frames, shown with their 95% confidence intervals. Triangles as in Fig. 5.

TABLE 1

Paleomagnetic results for the Miyako Cretaceous sediments

Site	N_s	D_g	I_g	α_{95}	D_s	I_s	α_{95}
JK 03	8	327	48.5	25	338	58	
JK 05	7	307.5	53	13	337	68	
JK 06	8	338.5	58.5	11	25	69	
JK 07	8	315	42	11	338	50.5	
JK 08	7	328	62	22	11	64.5	
JK 09	6	296	50.5	8	338	68	
JK 10	8	328.5	60	6	19.5	56	
JK 11	7	321	42	19	344	53	
Mean [a]	5	315.5	53.5	11.5	355	64	13
Mean [b]	8	319.5	52.5	7.5	351	62	8
HG 01	10	323.5	53	8.5	10.5	59.5	
HG 02	10	308.5	57.5	8	23	68.5	
HG 03	10	331.5	49.5	8	0	57.5	
HG 04	11	317	50.5	6.5	346	63.5	
HG 05	8	343	62.5	6.5	26	62	
TN 01	6	309	59	10.5	24.5	73.5	
Mean [c]	6	322	56	7.5	10.5	64.5	7.5
Mean [d]	14	320.5	54	5	0	63.5	5.5

Characteristic magnetization directions prior to bedding correction (D_g, I_g) and after bedding correction (D_s, I_s). N_s = number of specimens or sites (for means); α_{95} = 95% confidence oval. All angles in degrees.
Sites JK are from this paper, sites HG and TN from Tosha [1].
Descriptions of means: [a] mean of sites JK 05 to 07, 09 and 10; [b] mean of all sites; [c] mean of HG and TN sites; [d] mean of all JK, HG and TN sites.

TABLE 2

Compilation of pre-Cenozoic results from Northeast Japan

(a) Sedimentary rocks

Age	Location	N_S/N_s	D_g	I_g	α_{95}	Source
D	Kitakami	6	323.5	51	16	Tosha [1]
C	Kitakami	6	326	59	12	
P	Kitakami	8	313	48.5	16	
TR	Kitakami	8	331.5	48.5	10.5	
J	Kitakami	10	339.5	43.5	9	
K_1	Kitakami	1/6	330	59	12	site JK 01; this paper
Mean [a]		5	328.5	54.5	10.5	

(b) Plutonic rocks

Location	N_S/N_s	D_g	I_g	α_{95}	T	Source
South Kitakami	10	319	50	9	N	Kawai et al. [8]
South Kitakami	16	310.5	48	8	Y	Ito et al. [11]
Abukuma	3	323.5	60.5	9	N	Kawai et al. [8]
	4	320.8	50.6	5	Y	Ito and Tokieda [9]
Ashio	2	337	60.5	1	Y	Otofuji et al. [2]
Mean [b1]	5	321	54	8		
Mean [b2]		314	50	14		

(c) North Kitakami

Location	N_s	D_g	I_g	α_{95}	Source
Plutonic rock:					
Sugo	5	299	65	17	Kawai et al. [8]
Ochiai	6	326	34	24	
Hosozawa	7	333	35	19	
Otomo	4	331	58	20	
Yamaya	12	325	53	8	
Ichinohe	8	288	47.5	6	
		101	−30.5	20	
Ichinohe	8	109	−10	10	Ito et al. [11]
Numabukuro	8	288	26	8.5	
Kosode-Kaigan	10	277	31	10	
Kosode-Kaigan	8	258	−17	4	
Shimoakka	12	280	16	13.5	
Otanabe	10	268.5	0	5	
Kurumihata	9	300	20	7.5	
Uge	6	20	50	9	
Okawa	11	275	47	7.5	
Mean [c1]	5	325	49.5	15	
Mean [c2]	8	283	20.5	12	

A compilation of characteristic magnetization directions for sedimentary and plutonic rocks of pre-upper Cretaceous age in Northeast Japan, mostly from the Kitakami area. N_S: number of sites, N_s: number of specimens, T: cleaning techniques (N: simple AF cleaning, Y: stepwise AF and thermal cleaning). Sediments show a negative fold test (Tosha [1], and site JK 01 from this paper). For the means, the number of studies/formations which are averaged is indicated.

[a] The mean does not include JK 01, although this site is consistent with the rest.

[b1] Mean of five individual studies of plutons; [b2] the direction predicted by the Kitakami pole of Sasajima [4].

[c1] Mean of five sites studied by Kawai et al. [8]; [c2] mean of eight most significant sites studied by Ito et al. [11].

granitic massifs. This review starts with the oldest strata and ends with Miocene and Pliocene formations, and emphasizes evidence for or against the primary nature of the characteristic magnetizations isolated by various authors.

3.1. Devonian to Jurassic

In a very thorough study of Devonian to Jurassic sediments around the Kitakami massif, south of the Hayachine tectonic belt, Tosha [1] demonstrated that all these formations had very similar in-situ characteristic magnetizations (Fig. 6-1 and Table 2). Very clear negative fold tests were obtained on Jurassic, Triassic, Permian, Carboniferous and Devonian samples, for a total of 38 sites. The corresponding mean remagnetizations are not statistically distinct and provide an overall mean direction $D = 328°$, $I = 49.5°$ ($\alpha_{95} = 5.5°$, $N = 38$ taking sites as statistical units) or $D = 328.5°$, $I = 54.5°$ ($\alpha_{95} = 10.5°$, $N = 5$ taking formations as statistical units).

3.2. Early Cretaceous granites (125–110 Ma)

Many studies have been devoted to Cretaceous granites in North Japan. They differ greatly in terms of the techniques used and the reliability of published results. For instance, early results of Kawai et al. [8] were obtained after a simple AF cleaning, whereas recent results of Otofuji et al. [2] involve various stepwise cleaning techniques and the determination of characteristic directions. A summary of results is given in Table 2 and illustrated in Fig. 6-2. Results coming from the South Kitakami, Abukuma and Ashio massifs [2,8,9,11] are very consistent, regardless of study or techniques used to determine a direction (Table 2b; we have not included the 92 Ma old Iide granite from Otofuji et al. [2] because of its very anomalous inclination). We find an average direction $D = 321°$, $I = 54°$ ($\alpha_{95} = 5°$, $N = 5$ studies), which is close to that given for pre-1981 results in Sasajima's [4] review ($D = 314°$, $I = 50°$, $\alpha_{95} = 14°$, calculated for a site at 40°N, 142°E). We have no clue as to the primary or secondary nature of this magnetization direction, and of course, as is most often the case with granites, we have no clue as to possible tectonic tilting of the various plutons. All these possibilities will have to be considered in the discussion.

Interestingly, results published by Kawai [8] and Ito et al. [11] from the north Kitakami granites are largely discrepant (Table 2c and Fig. 7). Kawai used older paleomagnetic cleaning techniques. More recently, Ito et al. [11] were able to obtain what they considered to be significant results only for 8 out of 25 sampled sites. Their success rate was 16 out of 18 sites for the south of the massif. This suggests that the northern granites may have suffered at least one magnetization event. The Miyako sediments have been deposited over these northern granites and are presently tilted some 20° to the east. There is therefore a possibility that the underlying granites have also been tilted by a similar amount. We could assume that the component isolated by Kawai was a remagnetization, and that because they used more sophisticated techniques, Ito et al. [11] were able to better isolate a primary component. It so happens that if this component is corrected for the tilting of the Miyako series, it moves significantly towards the Kawai component $D = 325°$, $I = 49.5°$, $\alpha_{95} = 15°$ (which happens to be essentially identical both to the southern granites in-situ component and to the Miyako in-situ magnetization). These coincidences will be discussed in the next section. Of course the structural correction for the granites is only approximate and very tentative. The point of this discussion is to emphasize both the lack of proper structural control in the granites, and the definite possibility that at least some (and as we will see

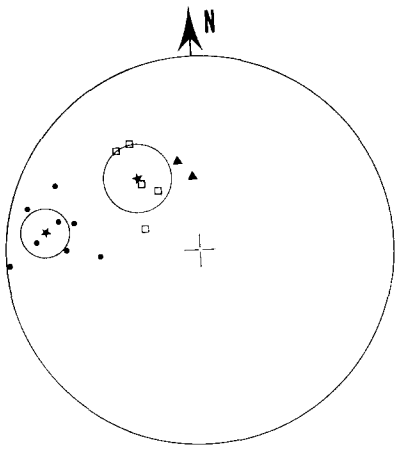


Fig. 7. A comparison of directions found in the Kitakami granites north of the Hayachine tectonic belt by simple AF cleaning [8] (squares) and by stepwise AF and thermal cleaning [11] (solid dots). Stars are averages and are shown with the α_{95} confidence interval.

TABLE 3

Compilation of paleomagnetic results for the 20–40 Ma period in Northeast Japan

Age (Ma)	Location	Ref.	N_s	P	D_s	I_s	α_{95}	Source
	North Asahi	6						
21	Higashi		12	R	332.5	50.5	2.4	Otofuji et al. [2]
(fission track)	Ohotorigawa							
	Oga Peninsula	3						
20–22	Daijima		6	N	330.9	57.1	2.6	Otofuji et al. [2]
(fission track)	Daijima		11	N	328.4	40.9	3.6	
	Daijima		5	N	315.6	54.5	1.8	Tosha [1]
	Daijima		5	N	297.0	58.2	6.4	
	Daijima		10	R	297.6	45.3	6.7	
	Daijima		5	N	318.2	44.3	13.9	
	Yuzawa	4						
22–25	Hatamura		10	R	312.7	33.8	3.6	Otofuji et al. [2]
(K-Ar)	Hatamura		11	R	314.9	73.6	3.5	
	Hatamura		6	R	326.5	74.1	7.7	
	Hatamura		8	R	320.0	76.1	3.6	
25	Akatsugawa		10	R	307.4	48.4	2.7	
(K-Ar)	Akatsugawa		8	R	358.2	75.1	10.9	
	Akatsugawa		6	R	313.2	41.4	7.4	
	Akatsugawa		9	R	318.6	50.8	4.2	
	Akatsugawa		8	R	323.1	54.7	8.1	
	Akatsugawa		12	R	322.2	53.1	2.0	
> 25	Yunosawagawa		9	N	303.9	67.9	2.7	
(K-Ar)	Yunosawagawa		9	N	318.2	48.5	1.9	
	Yunosawagawa		10	N	328.2	72.9	2.7	
	Ninohe	1						
32	Nisatai		10	N	292.8	39.8	2.3	Otofuji et al. [2]
(fission track)								
	Oga Peninsula	3						
25–40	Monzen		9	R	331.6	45.9	6.2	Tosha [1]
(K-Ar,	Monzen		5	R	292.3	56.2	10.4	
fission track)	Monzen		4	R	344.4	58.0	7.7	
	Monzen		5	R	317.0	50.5	7.1	
	Monzen		6	R	329.3	58.0	2.6	
	Monzen		4	R	315.6	47.7	14.2	
Mean			27		317.5	55.5	5.0	$K = 32.4$

Ref. refers to location in Fig. 2a. N_s = number of specimens or sites (for mean); P = polarity of the characteristic direction.

later maybe all) of the granites have been remagnetized.

3.3. Early Cretaceous Miyako sediments (110–100 Ma)

The next question to be answered pertains to the primary or secondary nature of the Miyako magnetization. Tosha [1] argues that it is primary, because of the lack of evidence for any thermal or chemical remagnetization event and because of a positive conglomerate test; this test is based on only four pebbles from the lowermost part of the Miyako Group, where it unconformably overlies the older 110–125 Ma old Kitakami granites.

Should this magnetization be primary, its age would then lie within the Long Normal Cretaceous superchron, which would be in good agreement with the fact that all samples we measured had a normal magnetization.

However, Tosha [1] does not apply the conglomerate test in the conventional manner. This test amounts to finding random directions of magnetization in pebbles *from* an underlying formation found *in* an overlying conglomerate. From

this, one may infer that the magnetization of the rocks *from which* the pebbles were derived could be primary, *not* that the rocks forming the matrix of the conglomerate, or other overlying rocks, have not been remagnetized. Therefore, Tosha's observation might favour the fact that the underlying strata from which the pebbles were derived (possibly the north Kitakami granites, although this is not specified by Tosha) have not been remagnetized. It does not tell us that the Miyako series was not remagnetized. The demagnetization characteristics of the Miyako samples described in this paper, the observation that sulfides may be the main carrier of magnetization, and the absence of any fold test certainly do *not* allow us to rule out the possibility that the formation is remagnetized. For instance, although this is admittedly a weak indication, we find that the various Miyako means of Tosha and ourselves become further apart and their α_{95} confidence intervals slightly increase upon tilt correction. In particular, the Tosha 6-site mean and our 8-site mean are not anymore in the intersection of their α_{95} confidence intervals, dropping from A to C quality classification in the scheme of Tauxe et al. [14, fig. 3].

3.4. Paleocene and Eocene (65–40 Ma)

New paleomagnetic data for formations of Cenozoic age from Northeast Japan have recently been published independently by Tosha [1] and Otofuji et al. [2]. A summary of earlier data has also been given by Sasajima [4].

Tosha's [1] results pertain to formations from the Oga peninsula (Fig. 2), spanning the period from 62 to 6 Ma (dated by K-Ar and fission tracks). These results are summarized in Table 3. For 9 specimens from the adamellite basement (62 Ma), the α_{95} confidence interval barely contains the present dipole field direction. The same is true for the unconformably overlying Akashima formation (53 Ma), which consists of welded tuffs and andesites. Otofuji et al. [2] have also studied this formation more extensively, and find a clearly negative fold test, the direction prior to bedding being identical to the present axial dipole field direction. Thus, the basement and Akashima formations have apparently suffered recent complete remagnetization. Otofuji et al. [2] also list results from three sites in 60 Ma old granodiorites. Although they are very dispersed ($\alpha_{95} = 35°$), the mean direction is close to that of the general remagnetization which affects pre-125 Ma old rocks.

3.5. Oligocene (40–20 Ma)

In the Oga peninsula, the Akashima formation is in turn unconformably covered by the Monzen basalts, andesites and rhyolites, dated from 40 to 25 Ma. Tosha [1] obtains a positive fold test (at the 95% level) for six sites sampled in this volcanic formation and a prefolding paleomagnetic direction $D = 322°$, $I = 54°$ ($\alpha_{95} = 9.5°$). Although no fold test is available for the overlying Daijima basalts and tuffs, dated from 25 to 20 Ma, Tosha [1] finds evidence for a reversal, and a significant westward declination, similar to the Monzen direction after bedding correction. Otofuji et al. [2] have also studied 2 sites of dacite welded tuffs from the Daijima formation and find a direction which is similar to that of the Monzen formation of Tosha [1] (apparently given after bedding correction by Otofuji et al., although the bedding parameters or directions prior to bedding correction are not given). The overall mean of Otofuji et al. [2] for 17 sites dated from 20 to 32 Ma ($D = 319°$, $I = 56°$, $\alpha_{95} = 7°$) is virtually identical with the Monzen direction of Tosha [1]. The results of Otofuji et al. [2] come not only from sites within the Oga peninsula, but actually cover a wide part of Northeast Japan, including Kitakami, North Asahi and Ninohe (Fig. 2). Because of the good agreement of all the 20 to 40 Ma studies and of the positive fold test for the Monzen formation, we conclude that the average direction for 27 sites ($D = 317.5°$, $I = 55.5°$, $\alpha_{95} = 5°$; Table 3) is likely to represent a primary field direction for Northeast Japan over that period of time. Should it correspond to a remagnetization, this remagnetization must have occurred prior to the folding of the Monzen formation.

3.6. Miocene to Present (20–0 Ma)

All published results for formations younger than 20 Ma show directions which are very different from Monzen, and are close to the presently expected axial dipole field direction. This is true for the ≈ 17 Ma Nishikurosawa, ≈ 12 Ma Onnegawa and ≈ 6 Ma Kitaura and Kampuzan formations in Oga peninsula (Table 3 and Tosha [1]).

TABLE 4

Compilation of paleomagnetic results since 20 Ma

Age (Ma)	Location	Ref.	N_s	P	D_s	I_s	α_{95}	Source
	Oga Peninsula	3						
17	Nishikurosawa		7	N	3.9	62.2	8.6	
(K-Ar,			7	N	−0.6	61.1	12.6	
fossils)			5	N	5.8	57.5	10.6	
			7	R	−10.8	48.1	11.3	
17	Oga dike swarm		5	R	17.1	59.7	19.7	
			5	N	1.6	58.5	8.4	
			4	N	−42.2	60.8	16.1	
			5	N	10.4	70.2	15.7	
			4	N	28.3	53.2	13.7	Tosha [1]
			5	R	18.7	38.4	19.3	
			4	R	−7.3	68.6	11.8	
			4	R	−25.6	53.7	11.3	
			4	N	−18.8	45.8	12.2	
			4	N	−26.1	42.9	6.2	
			5	N	−2.2	56.6	7.6	
			5	R	25.8	62.6	13.2	
			4	N	−10.7	62.6	6.0	
	Taihezan	2						
14	Okuramata		18	N	−10.6	42.9	4.7	Otofuji et al. [2]
(K-Ar)			10	N	−9.9	61.1	5.1	
	Zao	5						
11.3			12	R	−14.5	66.9	3.3	Otofuji et al. [2]
(K-Ar)	Yamadera		11	R	−13.5	71.8	1.9	
	Oga Peninsula	3						
12	Onnagawa		6	R	−7.6	60.0	10.7	
(fossils)			6	N	4.6	63.5	11.6	
			6	N	16.3	62.3	5.6	Tosha [1]
	Kampuzan		4	N	7.7	72.8	9.7	
6	Kitaura		4	N	12.4	37.8	17.7	
(K-Ar,			3	N/R	8.4	54.1	17.6	
fossils)								
Mean			27		−1.5	58.7	4.6	

Ref. refers to location in Fig. 2a. N_s = number of specimens or sites (for mean); P = polarity of the characteristic direction.

We note actually that both the Nishikurosawa and Onnegawa formations have significantly larger scatter after bedding correction, suggesting that some amount of recent viscous magnetization remains. However, the mean direction is hardly changed by the bedding correction. In much the same way, the 11 and 14 Ma old Yamadera dacite and Okuramata rhyolite welded tuffs [2] yield an average direction $D = 348°$, $I = 60.5°$ ($\alpha_{95} = 15°$), not far from the present dipole field. We can average results for 14 sites covering the period 0–17 Ma ([1,2]; and Table 4) and find the presently expected axial dipole field direction at $D = 0°$, $I = 59°$ ($\alpha_{95} = 5.5°$). This result is unchanged if we include results from 13 sites sampled by Tosha [1] in the Oga peninsula dike swarm, which belongs to the same period: $D = -1.5°$, $I = 59°$ ($\alpha_{95} = 4.5°$, $N = 27$ sites).

4. Synthesis of paleomagnetic data and geodynamic scenarios

The previous review shows that the paleomagnetic data set from Northeast Japan is more limited than is generally realized, and that the lack of sufficient stability tests still precludes one from

assigning a definite primary or secondary origin to several key paleomagnetic directions. Moreover, there is an intriguing set of coincidences such that most corrected or uncorrected directions happen to be virtually identical (Table 5, Fig. 6-6). In any case, there are no primary directions prior to 125 Ma. We have seen that all earlier rocks had suffered complete remagnetization. This remagnetization is identical to the in-situ direction of the 110–25 Ma old granites of South Kitakami, Abukuma and Ashio. It is also identical with various mean directions from granites compiled by Kawai et al. [3], Sasajima [4] and Tosha [1] (see Tables 2b and 2c). Therefore, we concur with Tosha in thinking that these directions are the same and are linked to the same remagnetization event. Tosha [1] concludes that the granites were responsible for the thermal remagnetization of all the pre-Cretaceous sediments which they intruded and that the corresponding overall mean gives a valid estimate of the field direction in Northeast Japan in the Lower-Cretaceous (110–125 Ma).

However, we have seen in the previous discussion that there was no compelling evidence for the primary nature of the granite magnetization. Also, there is no reason why the granites could not have been tilted somewhat posterior to their magnetization. For instance, in the northern part of the Kitakami massif, the Miyako series unconformably overlies the granites and must have been deposited flat upon them, so that they must have undergone similar subsequent tilting. It is not clear whether granites on either side of the Hayachine tectonic belt have suffered the same deformation. We have seen in the discussion that there was some evidence that two distinct components of magnetization had been isolated by separate authors in north Kitakami, whereas a single one was found in the south. The mean direction of the southern granites (Table 2b) happens to be identical to what could be a softer remagnetization in the northern granites [3] (Table 2c), but it is also similar to the harder component found by Ito et al. [11], once it has been tilted according to the Miyako sediments bedding (Fig. 7).

Also, the in-situ direction of the Miyako sandstones is statistically identical to *all* previously discussed directions (Fig. 6-6). Recall that we have no definite clue as to the primary nature of the Miyako magnetization itself. It could equally well be primary and identical to the primary magnetization of the granites, both having been tilted by an equal amount, or all could be remagnetizations posterior to tilting of the Miyako series. This would require block-tilting of the granite massifs along as yet unobserved faults, which does not seem impossible.

The next (younger) data are those with an age range of 20–40 Ma, which include the Monzen formation, the only one to display a clear positive foldtest; they lead to a paleodirection $D = 321.5°$, $I = 54°$ ($\alpha_{95} = 9.5°$). After bedding correction, this correction happens to be *identical* to that of the Miyako series prior to bedding correction. Finally, all rocks younger than 20 Ma (actually $\approx$ 17 Ma) have been magnetized in a direction which is not distinguishable from the present axial dipole field direction ($D = -1°$, $I = 59°$, $\alpha_{95} = 4.5°$, $N = 27$ sites from Tosha and Otofuji et al.).

In summary, only Eocene to Present data from Northeast Japan appear to be clearly primary in nature. The magnetization of the Cretaceous sedi-

TABLE 5

Available directions from North Japan

Age	In situ (if remagnetization)				Tilt corrected (if primary)			
	$D/\delta D$	$I/\delta I$	$\lambda/\delta\lambda$	α_{95}	$D/\delta D$	$I/\delta I$	$\lambda/\delta\lambda$	α_{95}
Lower Cretaceous (110–125)	323.5/24	54/10	34.5/12.5	10				
Miyako (100–110)	320.5/12	54.5/5	35/6.2	5	0/13.2	63.5/5.5	45/6.9	5.5
Oligocene					321.5/22.8	54/9.5	34.5/11.9	9.5
Miocene to Present					359/12	59/5	39.8/6.2	5

D, δD = declination and 95% error in D; I, δI = inclination and 95% error in I; λ, $\delta\lambda$ = paleolatitude and 95% error in λ.

TABLE 6

Various possibilities for the age of magnetization of Cretaceous rocks from North Japan

		Miyako sediments	
		primary magnetization, then tilted	remagnetization after tilting
Kitakami granites	primary magnetization, tilted	*Scenario 3* — two coincidences	— not very different from scenario 3 — not very plausible — implies 1 coincidence
	primary magnetization, not tilted	— considered unlikely — two coincidences and APWP hairpins [1]	*Scenario 2* — no coincidence — does not agree well with Eurasian APWP
	remagnetized	— not very different from scenario 3 — not very plausible (magnetite remagnetized, but not sulfides) — implies 2 coincidences	*Scenario 1* — no coincidence — cause of widespread remagnetization unknown

See text for discussion.

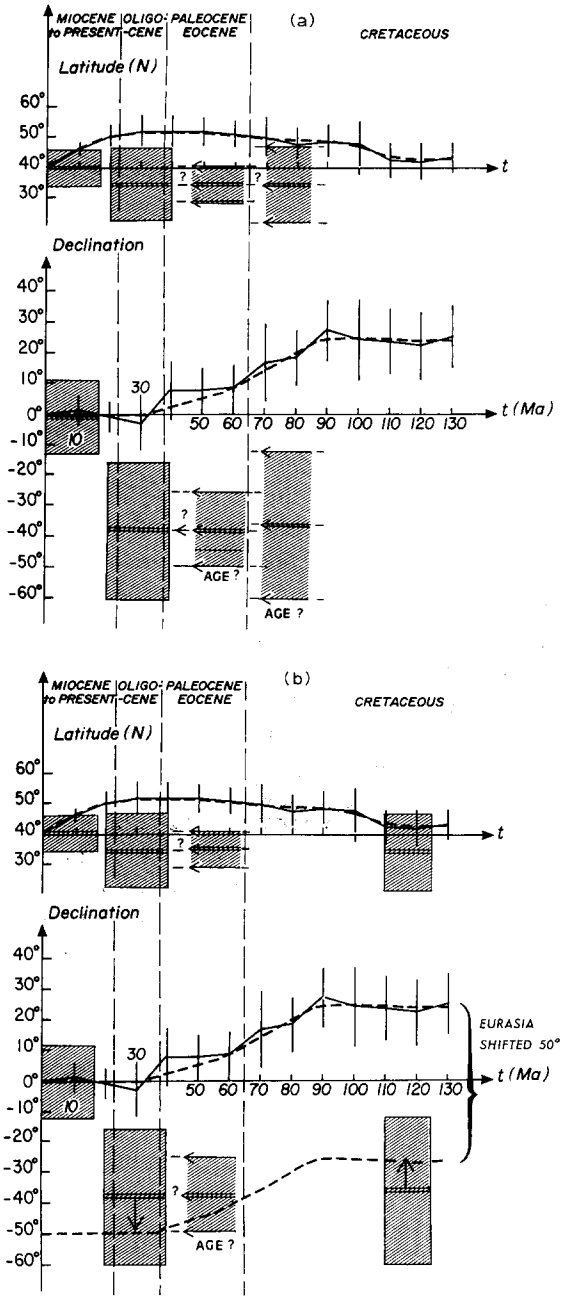
ments could be either primary (and therefore should be corrected for tilting) or secondary. The magnetization of the Cretaceous granites could be either primary with no tilting (as suggested by data south of Hayachine), or primary and tilted (as suggested by data north of Hayachine), or even completely remagnetized at some later time. Combining these possibilities gives rise to six different paleomagnetic, and therefore geodynamic histories for Northeast Japan (Table 6). We will now show that only three of those are actually reasonable. In order to make the discussion easier to follow, we have plotted the declination and paleolatitudes suggested for each scenario as a function of time. These data are compared to those predicted (at a reference point 40°N, 140°E) from the latest determination of the apparent polar wander path (APWP) of Eurasia [15,16].

4.1. Scenario 1: Miyako and granites are remagnetized

This may be the simplest, however distasteful, hypothesis. This remagnetization would have occurred after both Miyako and the granites were in their present positions, and therefore the in situ directions should be used in that case. The remagnetization could have occurred at any time after the (unknown) tilting of the Miyako formation. Indeed, the magnetization of the Monzen formation is not constrained to be primary, but only prior to folding. This folding must be completed before ≈ 20 Ma, a date after which all magnetizations from Northeast Japan are not rotated. Therefore, the age of the remagnetization could be younger than 40 Ma, and possibly as young as 20 Ma. In this hypothesis, there would remain no paleomagnetic information older than 40 (or 20?) Ma in Northeast Japan! This would readily account for the remarkable coincidence of in-situ directions from pre-Jurassic rocks, Cretaceous granites, Cretaceous sediments and tilt-corrected directions from Oligocene rocks (Fig. 8a). In this hypothesis, an overall mean remagnetization direction $D = 323°$, $I = 52°$ ($\alpha_{95} = 5°$, $N =$ 13 studies) could be derived from all the results listed in Tables 1 and 2. On the other hand, this scenario would require a mechanism for the complete low-temperature chemical remagnetization of diverse rock types with very different magnetic mineralogies. Although such widespread remagnetizations are not yet fully understood, there is now ample evidence for their existence over large areas at various times (see for instance Irving and Strong [17] for North America; Courtillot et al. [18] for Hercynian Europe; Lee et al. [7] for South Korea; [19]). Our own unpublished results on Cretaceous and Triasic samples from Southern Japan (Shikoku) show unexpected complete remagnetization in the present field of rocks from high-quality fresh road cuts, with no apparent indication of alteration or thermal events [20].

4.2. Scenario 2: Miyako is remagnetized—granites are primary and not tilted

For reasons already discussed, it is easy to envision how the Miyako sulfide-bearing sediments could have been remagnetized. As outlined above it is more difficult, yet not impossible, to imagine that the magnetite-bearing granites have



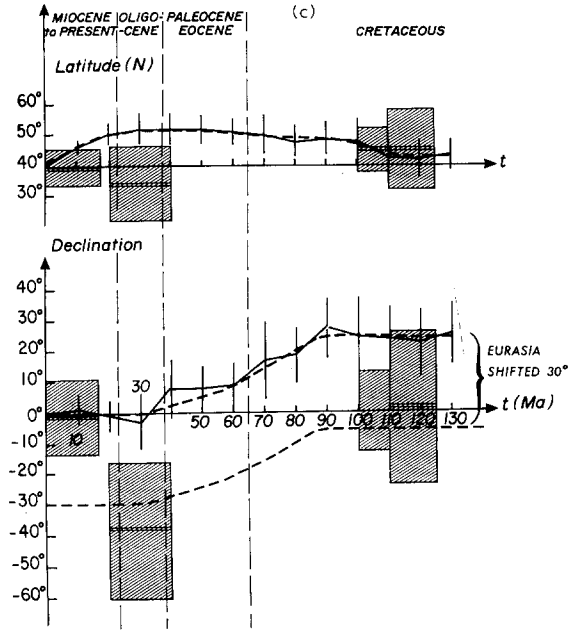


Fig. 8. Comparison of paleomagnetic directions (actually paleolatitude and declination D for a reference site at 40° N, 140° E) from North Japan with directions predicted for stable Eurasia [15,16] under various assumptions: (a) Scenario 1, in which it is assumed that the Miyako and Cretaceous granite directions are remagnetizations (in-situ directions) acquired after tilting of the Miyako series. Oligocene and Miocene directions have positive fold tests and are therefore given after bedding correction (see text and Tables 1–4). (b) Scenario 2, in which the magnetization of the Cretaceous granites is assumed to be primary, with no subsequent tilting of the massif. The predicted Eurasian declination curve is shifted by 50° for comparison with North Japan results. Otherwise as in (a). (c) Scenario 3, in which both the Miyako and Cretaceous granite directions are assumed to be primary and subsequently tilted. The predicted Eurasian declination curve is shifted by 30° for comparison with North Japan results. Otherwise as in (a).

also been remagnetized. However, should they both be primary and yet not tilted since cooling, as might be suggested by studies south of the Hayachine tectonic belt, another history could be proposed (Fig. 8b). We can assume that from prior to 125 Ma to about 20 Ma, Northeast Japan was part of the Eurasian continent. In-situ inclination of the granites agrees well with that predicted for Eurasia at that time (Fig. 8b). There is a declination discrepancy of 61° (±26°), which is larger but of the same sense as the declination discrepancy for the Oligocene results (35° ± 25°). The smooth predicted Eurasian declination curve can indeed be shifted to pass through the Northeast

Japan values within all 95% error bars (Fig. 8b). The mean declination difference that provides the "best fit" is about 50°. The discussion of the age of the Miyako remagnetization in this scenario is identical to that of the previous scenario. In slight disfavour of this scenario is the fact that all average directions are so perfectly coincident (Figs. 6-6 and 8b), which here would have to be fortuitous, considering the significant changes in directions predicted for Eurasia between the Cretaceous and the Oligocene ($D \approx 26° \pm 14°$). Also, this would fail to account for the discrepant observations of Kawai et al. [8] and Ito et al. [11] north of the Hayachine tectonic belt.

4.3. Scenario 3: Miyako and granites are primary and tilted

If we assume that both Miyako and the granites have a primary magnetization *and* that the granites are tilted in-situ directions to that of Miyako, then the coincidence of in-situ directions is preserved for tilt-corrected directions and is thus simply accounted for. The field direction would not have changed between 125–110 Ma (granites) and 110–100 Ma (sediments), which is reasonable. Moreover, the tilt corrected inclinations would agree remarkably well with those predicted for Eurasia at that time (Fig. 8c). Also, the declination change between the Cretaceous and the Oligocene would nicely reflect the corresponding change in Eurasian declination (Fig. 8c), with a mean difference of about 30°, i.e. significantly smaller than in scenario 2. This scenario, however, implies two fortuitous coincidences: first, the (uncorrected) primary directions of both Miyako and Kitakami would happen to be identical to the primary Oligocene direction, and second the (corrected) directions happen to be identical to the present field direction in North Japan.

4.4. Other scenarios

The three remaining possibilities (Table 6) imply even more implausible coincidences, or do not bring much new insight compared to the three previous ones. For instance, we can assume that the Miyako magnetization is primary and that of the granites secondary. This still implies two coincidences, namely that the corrected Miyako direction is identical to present field by chance, second that the primary Miyako in-situ direction is identical to the granite remagnetization and Oligocene primary magnetization by chance. Also, it is hard to see why the sulfide-bearing sediments would have been more resistant to remagnetization than the nearly magnetite-bearing granites. Finally, the geodynamical story implied from Cretaceous to Oligocene would essentially be identical to that predicted in scenario 3.

Should the granites have a primary and tilted magnetization, and the Miyako sediments be remagnetized, again the geodynamical story is very similar to that of scenario 3, and one fortuitous coincidence is required: the primary granite direction would happen to have been tilted precisely to the Miyako in situ remagnetization direction. Consequences of the Miyako remagnetization would be as in scenario 2.

The last possibility, i.e. that the Miyako magnetization is primary and has been tilted, and the magnetization of the granites is primary but they have not been tilted, is the one advocated by Tosha [1]. We find that it is the most unlikely solution, in that it requires the largest number of fortuitous coincidences and generates the most erratic APWP, the one most dissimilar to the Eurasian APWP. Indeed, this scenario would imply an APWP with two sudden U turns (hairpins) for North Japan: the lower Cretaceous (granite) and Oligocene (Monzen) poles would happen to be identical, and so would the Cretaceous (Miyako) and present-day poles. Apparent polar wander would have to be particularly fast around 110 Ma (see [1, fig. 7.3]). We find that the coincidence of in-situ magnetic directions from the nearby (both in space and time) granites and Miyako sediments cannot be simply ascribed to chance. In any case, logical ranking of solutions requires testing (and possibly falsifying) first those scenarios that require the smallest number of coincidences.

5. Conclusion—Miocene opening of the Japan Sea

The three possible histories described above share the same data for the last 40 Ma, which constrains the Miocene opening of the Jpan Sea. Estimates of the amount of rotation between Northeast Japan and the Eurasian landmass during the Miocene only weakly depend on the hypotheses made on for older periods data.

The rotation can first be estimated simply from

comparisons of 20–40 Ma and 0–20 Ma directions, and corresponding directions for Eurasia (Fig. 8). The declination difference is then 37° (±26°). Should one accept scenario 1, the overall remagnetization direction is known with a much better precision ($\alpha_{95} = 5°$) and the declination difference becomes 37° (±12°). Now, if one accepts scenario 2 (Fig. 8b), the various Cretaceous Oligocene paleomagnetic data can be combined with the Eurasian predicted curve and suggest a larger rotation of about 50°. If scenario 3 is preferred, the same reasoning (Fig. 8c) leads to a smaller rotation of about 30°. In any case, it seems that Northeast Japan cannot have rotated less than 20–30° counterclockwise when the Sea of Japan opened. This result critically relies on the quality of Oligocene results, and particularly on the only positive fold test from the Monzen formation [1]. It would favour rotational models of opening of the Japan Sea over translational models that propose a pure pull-apart mode of basin formation (e.g. [6]). Faure and Lalevée [5] have recently poposed a rotational or bending model based on the observation of trends of pre-Miocene structures in Japan. The rotation angle which they propose between North Japan and Shikoke Alin is about 35°, i.e. fully compatible with our discussion of paleomagnetic results. However, we must point out our unresolved discrepancy in all these models with respect to the paleomagnetic data; Fig. 8 shows a paleolatitude difference of 15° (±13°) between North Japan and Eurasia in the Oligocene, prior to the opening of the Sea of Japan. Although not well resolved from a statistical point of view, this discrepancy is of a sense opposite to that predicted by most models, which call either for minor relative motion, or for some amount of *southward*, but not *northward* motion (e.g. [5,6]). It therefore appears that present Oligo-Miocene poles for Eurasia do not accurately predict the paleolatitude of Northeast Japan, which is very consistently found to be about 35° N (for points presently located near 40° N), i.e. some 500 km south of its present latitude.

Recent compilations of paleomagnetic data from Japan have failed to recognize the insufficient constraints on the primary nature of magnetization for some key periods. For instance, the compilation of Celaya and McCabe [21] uses Tosha's [1] results at about 50 Ma and 65 Ma (their fig. 2): we have shown in the discussion that the first provides a clear negative fold test [2] and that the second has large error bars and is not statistically distinct from the present field. It is our feeling that published results from South Japan should be submitted to the same kind of critical review performed here for North Japan. Our unpublished finding that Cretaceous and Triasic rocks from Shikoku are completely remagnetized in the present field does argue for some caution in using published results with few (or no) stability tests and no indications of bedding corrections (see [19,20]). For instance, we doubt the validity of the single 60 Ma result listed by Celaya and McCabe [21] for South Japan. As a result, we feel that the pre-Oligocene stage of opening of the Japan basin proposed by these authors cannot at present be defended on the basis of paleomagnetic data. Moreover, this model fails to recognize changes in the position of Eurasia implied by its APWP [15,16]. In the same way, many recent models that rely too heavily on pre-40 Ma paleomagnetic data from Japan can no longer be regarded as valid.

In this paper, we have outlined three scenarios which are compatible with the data and require further testing. We believe that they should be analysed in order of increasing complexity and increasing demand for fortuitous coincidences. Therefore, we feel that the model that should be tested first, and should be considered as potentially valid until proven otherwise, is scenario 1, in which it is assumed that *all* rocks from North Japan may have suffered complete remagnetization as late 20 Ma. In this respect, determining the age of folding of the Monzen volcanics in the Oga peninsula would be an important constraint. It might be proposed that this widespread low temperature chemical remagnetization occurred just prior to the opening of the Sea of Japan, at the time where the Monzen and Daijima volcanics were erupted. This may correspond to a period of unusual tectonic, volcanic and thermal activity that marked the first stages of opening of the Sea of Japan. Kobayashi [22] states that magmatic activity in northeastern Japan appears to have been weak in the Paleogene and that it climaxed in the early Neogene, from 25 Ma on. The remagnetization could be related to the early phases of this active period. Recall that lower Cretaceous

sediments from the nearby Gyeongsang basin in South Korea have been extensively over-printed in recent times [7], a widespread and intense remagnetization which had previously failed to be identified [23].

Further work would require stability tests on older formations. It is probably no simple matter to further ascertain the primary nature of the Miyako sediments and Kitakami granites by additional fieldwork. Detailed structural and paleomagnetic work might first focus on Eocene rocks from the Oga peninsula and elsewhere in Northeast Japan. A similar program should be undertaken for formations from South Japan, again with careful structural tests. Until then, we are faced with the fact that the classical bending of the Japanese Islands rests on much weaker grounds than has been commonly realized and that we may not yet safely base a picture of the paleogeographic evolution of Japan prior to the Oligocene on paleomagnetic observations.

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